

FACULTY OF ENGINEERING
DIPLOMA ENGINEERING
DEPARTMENT OF INFORMATION TECHNOLOGY



BASICS OF OPERATING SYSTEM (DE104304)

QUESTION BANK

SECTION A: 3 MARKS QUESTIONS (25 Questions)

UNIT 1: Introduction to Operating System

Important 3 Marks Questions

1. Define an Operating System. Explain its fundamental goals.
 2. Differentiate between Multiprogramming and Time Sharing Operating Systems.
 3. Explain the services provided by an Operating System.
 4. What is a Real-Time Operating System? State its applications.
 5. Explain the generations of Operating Systems.
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UNIT 2: Process Management and Inter Process Communication

Important 3 Marks Questions

1. Define a Process. Explain the Process Life Cycle with a neat diagram.
 2. Explain the Process Control Block (PCB).
 3. Explain the FCFS CPU Scheduling Algorithm with an example.
 4. What is a Semaphore? Explain its role in process synchronization.
 5. List and explain the four necessary conditions for Deadlock.
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UNIT 3: Memory Management

Important 3 Marks Questions

1. Differentiate between Logical Address and Physical Address.
 2. Explain Internal Fragmentation and External Fragmentation.
 3. Explain the First Fit, Best Fit, and Worst Fit memory allocation techniques.
 4. Explain Paging with a neat diagram.
 5. What is Segmentation? State its advantages.
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UNIT 4: File Management System

Important 3 Marks Questions

1. Define a File. Explain various File Attributes.
 2. Explain different File Allocation Methods.
 3. Explain different Directory Structures in an Operating System.
 4. Explain the SCAN Disk Scheduling Algorithm.
 5. Differentiate between SCAN and C-SCAN Disk Scheduling Algorithms.
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UNIT 5: Linux and Shell Programming

Important 3 Marks Questions

1. Explain the basic architecture of Unix/Linux.
2. Write the syntax and uses of the following Linux commands: pwd, cd, ls, mkdir, and cat.
3. Differentiate between vi, vim, and gedit editors.
4. Explain the basic structure of a Shell Script with an example.
5. Explain the if statement and for loop in Shell Scripting with syntax.

SECTION B: 6 MARKS QUESTIONS (15 Questions)

UNIT 1: Introduction to Operating System

Important 6 Marks Questions

1. Explain the fundamental goals and services of an Operating System with a neat diagram.
 2. Compare Multiprogramming, Time Sharing, Real-Time, Multithreading, and Distributed Operating Systems.
 3. Explain the architecture and features of the Linux Operating System.
 4. Describe the generations of Operating Systems with suitable examples.
 5. Compare Linux and the latest Windows Operating System based on architecture, features, security, and applications.
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UNIT 2: Process Management and Inter Process Communication

Important 6 Marks Questions

1. Explain the Process Life Cycle with a neat diagram. Describe the role of the Process Control Block (PCB).
 2. Explain FCFS, Shortest Job First (SJF), and Round Robin Scheduling Algorithms with suitable examples.
 3. Discuss CPU Scheduling Criteria (CPU Utilization, Throughput, Turnaround Time, Waiting Time, and Response Time).
 4. Explain Inter Process Communication (IPC), Critical Section Problem, Semaphore, and Race Condition with examples.
 5. Explain Deadlock. Discuss its characteristics, prevention, avoidance, detection, and recovery techniques.
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UNIT 3: Memory Management

Important 6 Marks Questions

1. Explain Logical Address and Physical Address. Describe memory relocation and protection mechanisms with a neat diagram.
 2. Explain Contiguous Memory Allocation. Differentiate between Fixed and Variable Partitioning.
 3. Explain Internal Fragmentation, External Fragmentation, and Compaction with suitable examples.
 4. Explain First Fit, Best Fit, and Worst Fit Memory Allocation techniques with examples.
 5. Explain Paging and Segmentation with neat diagrams. Compare Paging and Segmentation.
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UNIT 4: File Management System

Important 6 Marks Questions

1. Explain the File System. Describe file attributes, file operations, and file types.

2. Explain different Directory Structures with neat diagrams.
 3. Explain File Allocation Methods (Contiguous and Linked Allocation) with their advantages and disadvantages.
 4. Explain Disk Structure and SCAN & C-SCAN Disk Scheduling Algorithms with suitable examples.
 5. Discuss File Protection mechanisms and Directory Protection in an Operating System.
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UNIT 5: Linux and Shell Programming

Important 6 Marks Questions

1. Explain the basic architecture of Unix/Linux with a neat diagram.
2. Explain the purpose and syntax of Linux commands (pwd, cd, mkdir, rmdir, ls, cat, cp, rm, mv, wc, grep, sort, head, tail) with examples.
3. Explain text editors in Linux (vi, vim, and gedit) and their features.
4. Write and explain Shell Programs using command-line arguments, logical operators, and the test command.
5. Write Shell Scripts to demonstrate the use of if, case, for, and while statements with suitable examples.

SECTION C: MULTIPLE CHOICE QUESTIONS (30 Questions)

UNIT 1: Introduction to Operating System

1. Which of the following is the primary goal of an Operating System?

- A) To increase hardware cost
- B) To manage computer resources efficiently
- C) To write application software
- D) To replace the CPU

✓ Answer: B) To manage computer resources efficiently

2. Which operating system allows multiple users to share CPU time simultaneously?

- A) Batch Operating System
- B) Time Sharing Operating System
- C) Single User Operating System
- D) Embedded Operating System

✓ Answer: B) Time Sharing Operating System

3. Which of the following operating systems is open source?

- A) Windows 11
- B) macOS
- C) Linux
- D) MS-DOS

✓ Answer: C) Linux

4. Which type of operating system is used in air traffic control and medical monitoring systems?

- A) Distributed OS
- B) Time Sharing OS
- C) Real-Time Operating System
- D) Batch OS

✓ Answer: C) Real-Time Operating System

5. Which generation of operating systems introduced multiprogramming?

- A) First Generation
- B) Second Generation
- C) Third Generation
- D) Fourth Generation

✓ Answer: C) Third Generation

UNIT 2: Process Management & Inter Process Communication

1. Which data structure stores information about a process?

- A) Stack
- B) Queue
- C) Process Control Block (PCB)
- D) Buffer

✓ Answer: C) Process Control Block (PCB)

2. Which CPU scheduling algorithm is non-preemptive?

- A) Round Robin
- B) FCFS
- C) Priority (Preemptive)
- D) Multilevel Queue

✓ Answer: B) FCFS

3. Which scheduling algorithm uses a fixed time quantum?

- A) SJF
- B) FCFS
- C) Round Robin
- D) Priority Scheduling

✓ Answer: C) Round Robin

4. Which synchronization mechanism is commonly used to solve the critical section problem?

- A) Cache
- B) Semaphore
- C) Stack
- D) Buffer

✓ Answer: B) Semaphore

5. Which of the following is one of Coffman's deadlock conditions?

- A) Mutual Exclusion
- B) Paging
- C) Segmentation
- D) Swapping

✓ Answer: A) Mutual Exclusion

UNIT 3: Memory Management

1. Which memory address is generated by the CPU?

- A) Physical Address
- B) Logical Address
- C) Virtual Disk Address
- D) Cache Address

✓ Answer: B) Logical Address

2. Which memory allocation technique selects the first available free partition?

- A) Best Fit
- B) Worst Fit
- C) First Fit
- D) Next Fit

✓ Answer: C) First Fit

3. Which type of fragmentation occurs in fixed partition memory allocation?

- A) External Fragmentation
- B) Internal Fragmentation
- C) Paging
- D) Swapping

✓ Answer: B) Internal Fragmentation

4. Paging is mainly used to eliminate:

- A) Internal Fragmentation
- B) External Fragmentation
- C) Deadlock
- D) Starvation

✓ Answer: B) External Fragmentation

5. Which memory management technique divides memory into logical units of variable size?

- A) Paging
- B) Segmentation
- C) Swapping
- D) Buffering

✓ Answer: B) Segmentation

UNIT 4: File Management System

1. Which of the following is NOT a file attribute?

- A) Name
- B) Size
- C) Creation Date
- D) CPU Speed

✓ Answer: D) CPU Speed

2. Which file allocation method stores files in consecutive disk blocks?

- A) Linked Allocation
- B) Indexed Allocation
- C) Contiguous Allocation
- D) Dynamic Allocation

✓ Answer: C) Contiguous Allocation

3. Which disk scheduling algorithm moves the disk head in one direction and then reverses?

- A) FCFS
- B) SSTF
- C) SCAN
- D) C-SCAN

✓ Answer: C) SCAN

4. Which directory structure allows files to be organized in multiple levels?

- A) Single-Level Directory
- B) Two-Level Directory
- C) Tree-Structured Directory
- D) Flat Directory

✓ Answer: C) Tree-Structured Directory

5. Which secondary storage device is commonly used for permanent data storage?

- A) RAM
- B) Cache
- C) Hard Disk
- D) Register

✓ Answer: C) Hard Disk

UNIT 5: Linux & Shell Programming

1. Which Linux command displays the current working directory?

- A) ls
- B) pwd
- C) cd
- D) mkdir

✓ Answer: B) pwd

2. Which command is used to create a new directory?

- A) mkdir
- B) rmdir
- C) touch
- D) ls

✓ Answer: A) mkdir

3. Which editor is commonly used in Linux for text editing?

- A) Notepad
- B) WordPad
- C) vi
- D) Paint

✓ Answer: C) vi

4. Which command is used to search for a pattern in a file?

- A) sort
- B) grep
- C) head
- D) tail

☒ Answer: B) grep

5. Which looping statement is available in Linux shell scripting?

- A) repeat
- B) while
- C) foreach only
- D) do-while only

☒ Answer: B) while